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# Install required libraries
!pip install langchain langchain_community cohere faiss-cpu pypdf

import langchain, cohere
print("LangChain Version:", langchain.__version__)
print("Cohere Version:", cohere.__version__)

!pip install langchain-cohere
from langchain.document_loaders import PyPDFLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain.vectorstores import FAISS
from langchain.chains import RetrievalQA
from langchain.embeddings import HuggingFaceEmbeddings
from langchain.llms import HuggingFacePipeline
from transformers import AutoTokenizer, AutoModelForCausalLM, pipeline
import torch

# ✅ STEP 1: Load IPC PDF
pdf_path = "10th class english 2020-21 20.pdf" # Ensure this file is
uploaded to Colab
loader = PyPDFLoader(pdf_path)
documents = loader.load()

# ✅ STEP 2: Split text into chunks
text_splitter = RecursiveCharacterTextSplitter(chunk_size=500,
chunk_overlap=100)
texts = text_splitter.split_documents(documents)

# ✅ STEP 3: Use Free Hugging Face Embeddings
embeddings =
HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-v2")

# ✅ STEP 4: Store embeddings in FAISS Vector Database
vector_store = FAISS.from_documents(texts, embeddings)

# ✅ STEP 5: Load Falcon-7B-Instruct (with Quantization to Save Memory)
model_id = "tiiuae/falcon-7b-instruct"

tokenizer = AutoTokenizer.from_pretrained(model_id)
model = AutoModelForCausalLM.from_pretrained(

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    model_id,
    torch_dtype=torch.float16,  # Efficient memory use
    device_map="auto"  # Automatically assign GPU/CPU
)

print("✅ Falcon-7B Loaded Successfully!")

# ✅ STEP 6: Create Text Generation Pipeline
pipe = pipeline(
    "text-generation",
    model=model,
    tokenizer=tokenizer,
    torch_dtype=torch.float16,
    device_map="auto",
    max_new_tokens=512,
    do_sample=True,
    temperature=0.7,
)

# ✅ STEP 7: Create Retrieval-based QA System
llm = HuggingFacePipeline(pipeline=pipe)
qa_chain = RetrievalQA.from_chain_type(llm,
retriever=vector_store.as_retriever())

print("Done all steps successfully")

# ✅ STEP 8: Chatbot Interaction
def chatbot():
    print("💎 IPC Chatbot is ready! (Type 'exit' to stop)")
    while True:
        query = input("You: ")
        if query.lower() == "exit":
            print("Chatbot: Goodbye! 🙋")
            break
        response = qa_chain.run(query)
        print(f"Chatbot: {response}")

# Run the chatbot
chatbot()

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